

SECTION 4: MOSFETS

ECE 322 – Electronics I

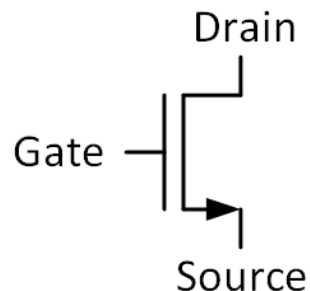
2 MOSFET Device Introduction

MOSFETs

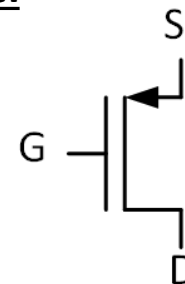
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- We now turn our attention to another type of transistor, the **MOSFET**:
 - ▣ **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor
- Many similarities to the BJT:
 - ▣ Three terminals
 - ▣ Voltage at one terminal controls current between the other two
 - A transconductance device
 - ▣ Two polarities: N-channel and P-channel MOSFETs
 - Our focus will primarily be N-channel MOSFETs (NMOS devices)

N-Channel (NMOS):



P-Channel (PMOS):

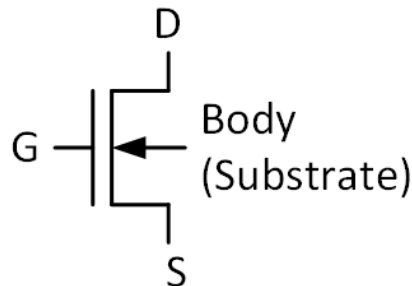


MOSFETs

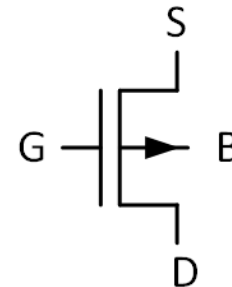
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- MOSFETs are actually ***four-terminal*** devices
 - ▣ Fourth terminal is the ***body, substrate, or bulk***
- The body is often tied to the source, and we can mostly ignore it
 - ▣ Discrete devices
- Other times we must account for the body potential effect on device behavior
 - ▣ Often the case in integrated circuits

NMOS:

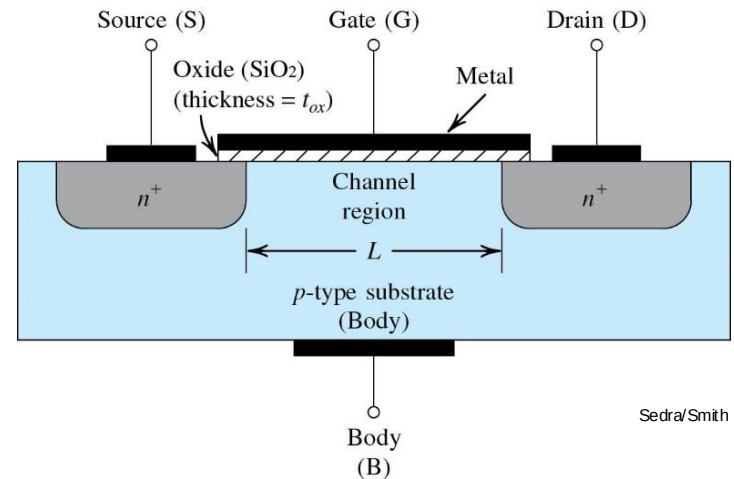
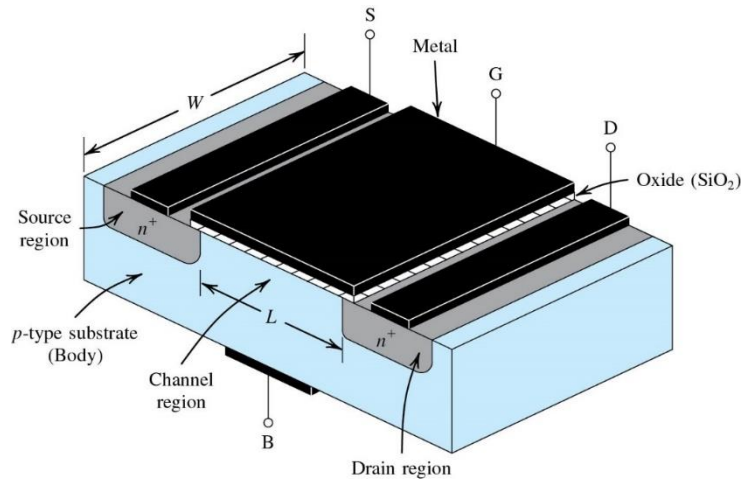


PMOS:



Physical Structure - NMOS

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- ❑ P-type substrate
- ❑ N+ source and drain
- ❑ **Metal gate electrode**, and source/drain/body contacts
- ❑ Thin **oxide** insulates the gate from the rest of the device
- ❑ Region of substrate between the drain and source is the **channel**
 - Channel dimensions: W and L
 - We'll see later why this is called an n-channel (NMOS) device

Terminal Voltages and Currents

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- Terminal voltages and currents named as shown
 - ▣ Again, lower-case v/i and upper-case subscript represents **total** (AC and DC) voltage and current
- For an NMOS device in typical operation:

$$v_{GS} \geq 0$$

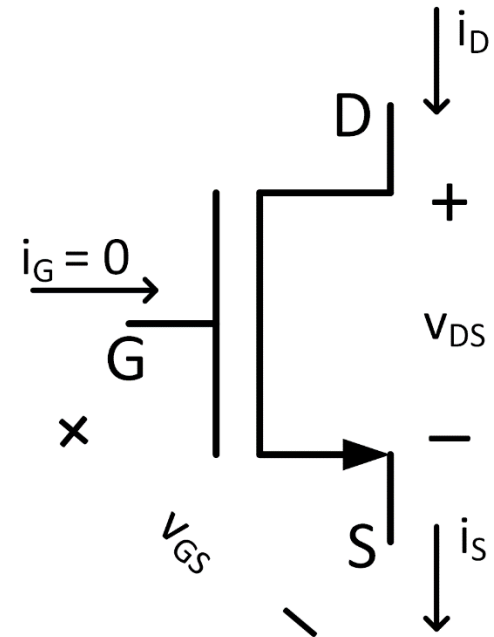
$$v_{DS} \geq 0$$

- Gate oxide does not allow current to flow, so

$$i_G = 0$$

and

$$i_D = i_S$$



MOSFET Operating Regions

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- Three MOSFET operating regions:
 - ▣ Cut-off
 - ▣ Triode
 - ▣ Saturation
- A MOSFET's operating region is determined by its terminal voltages
- Next, we will look in detail at each of these three regions, along with their $i - v$ characteristics

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Cut-Off Region

Cut-Off Region

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- Gate and source both grounded

$$v_{GS} = 0$$

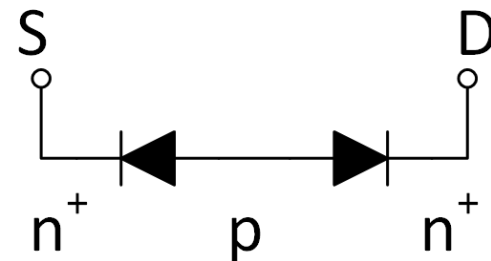
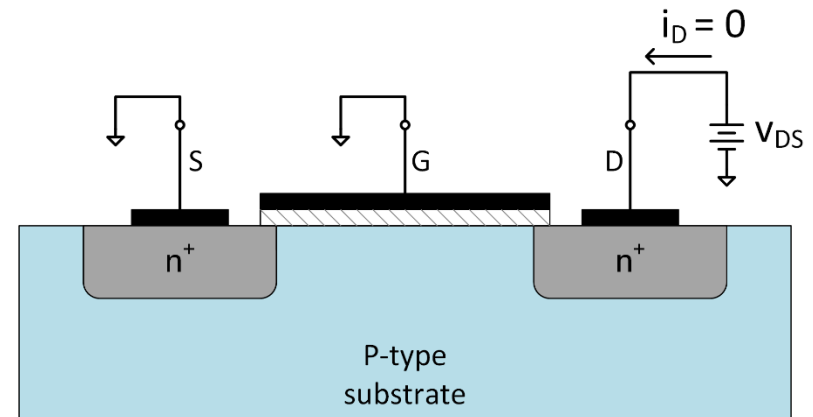
- Drain-to-source pathway looks like two back-to-back diodes

- Very high drain-source resistance ($r_{DS} = \infty$)

- Even for $v_{DS} > 0$, no current will flow

$$i_D = 0$$

- Looks like an open switch
 - Similar to BJT cut-off operation

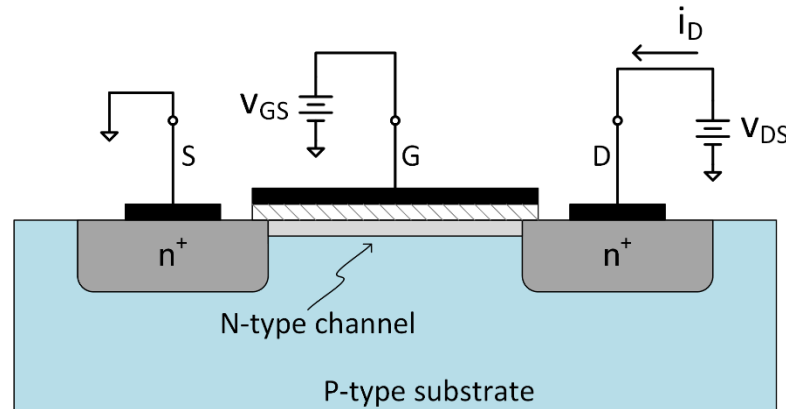


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Triode Region

Triode Region – Inversion

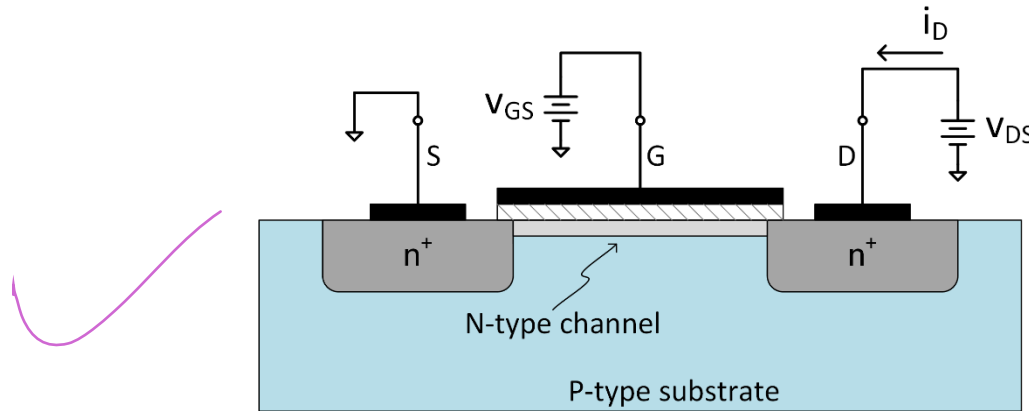
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- Now, v_{GS} is increased, while v_{DS} is kept small
 - ▣ Electric field established across gate oxide
 - ▣ Holes in p-type substrate repelled deeper into substrate
 - ▣ Electrons from drain and source attracted to region below the gate
- For large enough v_{GS} , p-type material below the gate is ***inverted*** to n-type
 - ▣ An ***inversion layer***
 - ▣ ***Induced n-type channel connects drain to source***
 - ▣ Now, current can flow in response to v_{DS} , $i_D > 0$

Threshold Voltage

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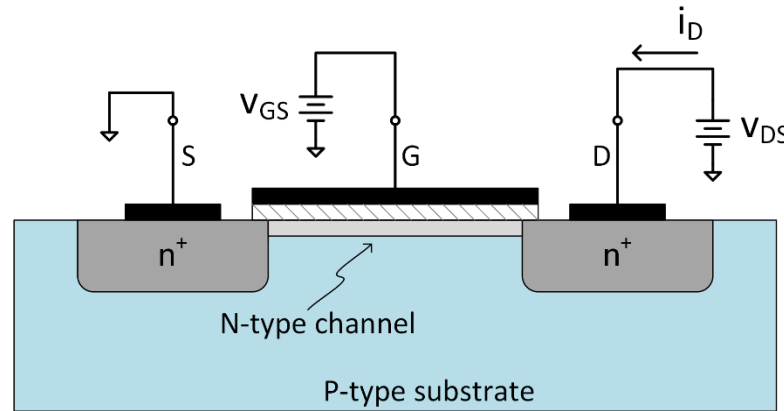
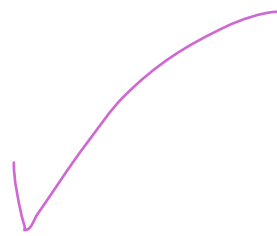
- Channel is induced once v_{GS} exceeds a certain voltage:
 - ▣ The **threshold voltage**

$$v_{GS} \geq V_t$$

- ▣ A device parameter
 - ▣ Typically, $V_t = 300 \text{ mV} \dots 1 \text{ V}$
- As v_{GS} increases beyond V_t , the induced channel gets deeper
- As long as v_{DS} is small ($v_{DS} \ll V_t$), channel depth is uniform

Overdrive Voltage

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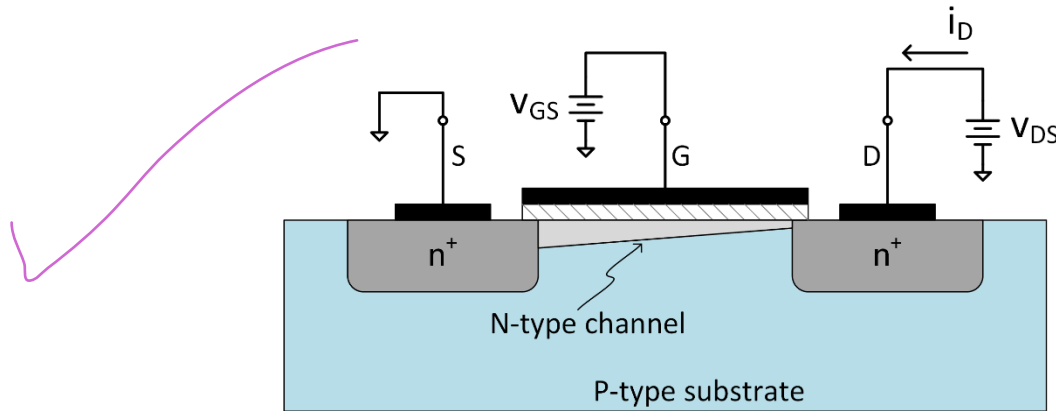
- A channel is induced once v_{GS} exceeds the threshold voltage
- v_{GS} in excess of the threshold voltage is called the **overdrive voltage** or **effective voltage**:

$$v_{OV} = v_{GS} - V_t$$

- As we will soon see, v_{OV} plays an important role in determining device behavior

Triode Region

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- As v_{DS} increases:
 - ▣ Voltage varies along the channel
 - v_S near the source, v_D near the drain
 - ▣ Gate-to-channel voltage decreases closer to the drain
 - ▣ Channel depth decreases closer to the drain
 - Channel is tapered
- More current flows with increasing v_{DS} , but channel resistance increases as channel becomes more tapered

Triode Region - i - v Relationship

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- Drain current in the triode region:

$$i_D = \mu_n C_{ox} \left(\frac{W}{L} \right) \left[(v_{GS} - V_t) - \frac{1}{2} v_{DS} \right] v_{DS}$$

$$i_D = \mu_n C_{ox} \left(\frac{W}{L} \right) \left[v_{OV} - \frac{1}{2} v_{DS} \right] v_{DS}$$

where:

- μ_n : electron mobility
- C_{ox} : oxide capacitance
- W : channel width
- L : channel length

$$K \left(2 (V_{OV}) V_{DS} - V_{DS}^2 \right)$$

- We can also express the drain current as

$$i_D = k'_n \left(\frac{W}{L} \right) \left[v_{OV} - \frac{1}{2} v_{DS} \right] v_{DS} = k'_n \left(\frac{W}{L} \right) \left[v_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

where:

- $k'_n = \mu_n C_{ox}$ is the **process transconductance parameter**

Triode Region - i - v Relationship

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□ Triode region:

$$v_{GS} > V_t$$

$$v_{DS} < v_{OV}$$

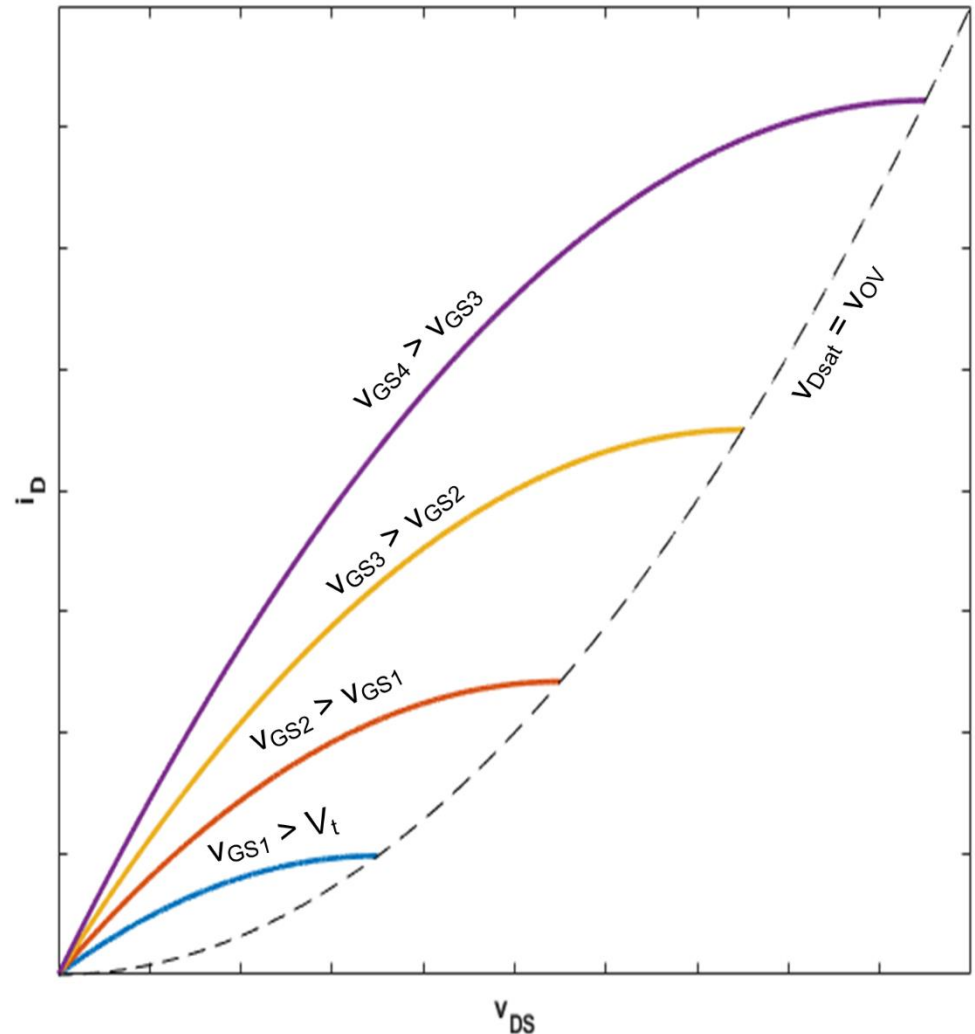
□ For $v_{DS} \ll v_{ov}$

- Nearly a linear resistance
- Resistance linearly proportional to v_{OV}

$$r_{DS} = \frac{1}{k'_n \frac{W}{L} v_{OV}}$$

□ As v_{DS} increases

- Channel taper increases
- r_{DS} increases
- i_D - v_{DS} slope decreases

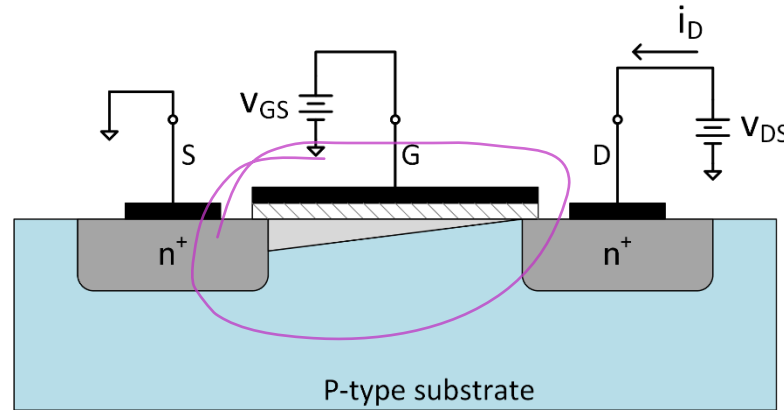


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Saturation Region

Device Operation – Channel Pinch-Off

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- Eventually, for large enough v_{DS}
 - ▣ Gate-to-channel voltage near the drain no longer exceeds V_t
 - ▣ Channel **pinch-off** occurs
 - ▣ Channel disappears at the edge of the drain
- Pinch-off occurs when:

$$v_{GD} = V_t = v_{GS} - v_{DS}$$

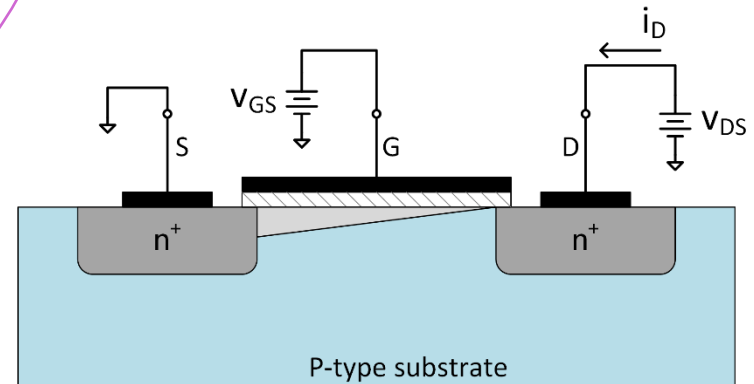
$$v_{DS} = v_{GS} - V_t$$

$$v_{DS} = v_{OV}$$

Saturation Region

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- Once channel pinch-off occurs:
 - Voltage at the drain-end of the channel remains v_{OV} , even as v_{DS} increases
 - Any increase in v_{DS} beyond v_{OV} is dropped across the depletion region surrounding the drain
 - Voltage across the length of the channel is fixed at v_{OV}
 - Pinched-off channel shape does not change with v_{DS}
 - Drain current **saturates** at a constant value for constant v_{GS}
 - Analogous to the forward-active region for BJTs



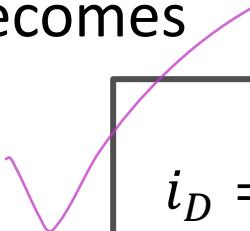
Saturation - i - v Relationship

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- Drain current in the saturation region still given by

$$i_D = k'_n \left(\frac{W}{L} \right) \left[v_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

- But now, the voltage from the drain-end to source-end of the channel is v_{OV}
- Replacing v_{DS} with v_{OV} , the drain current relationship becomes


$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2 = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_t)^2$$

- Purely a function of v_{GS} (or v_{OV})
- Independent of v_{DS}

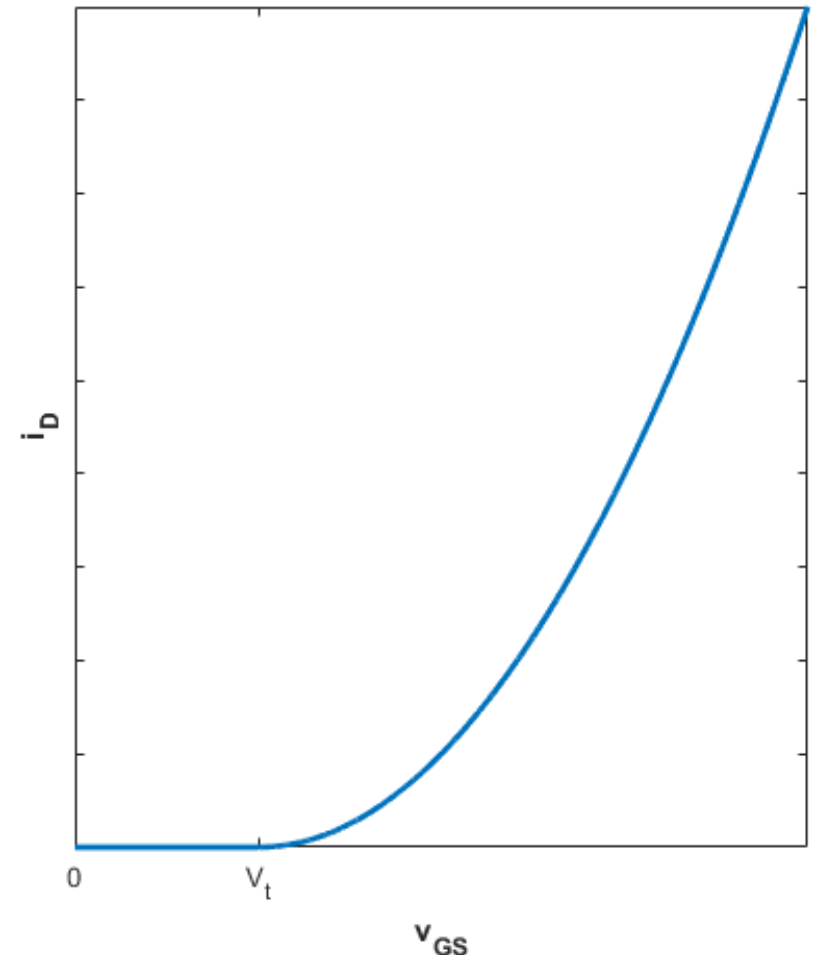
Input I-V Characteristic

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- In saturation, drain current has a **quadratic** dependence on v_{GS} (v_{OV})

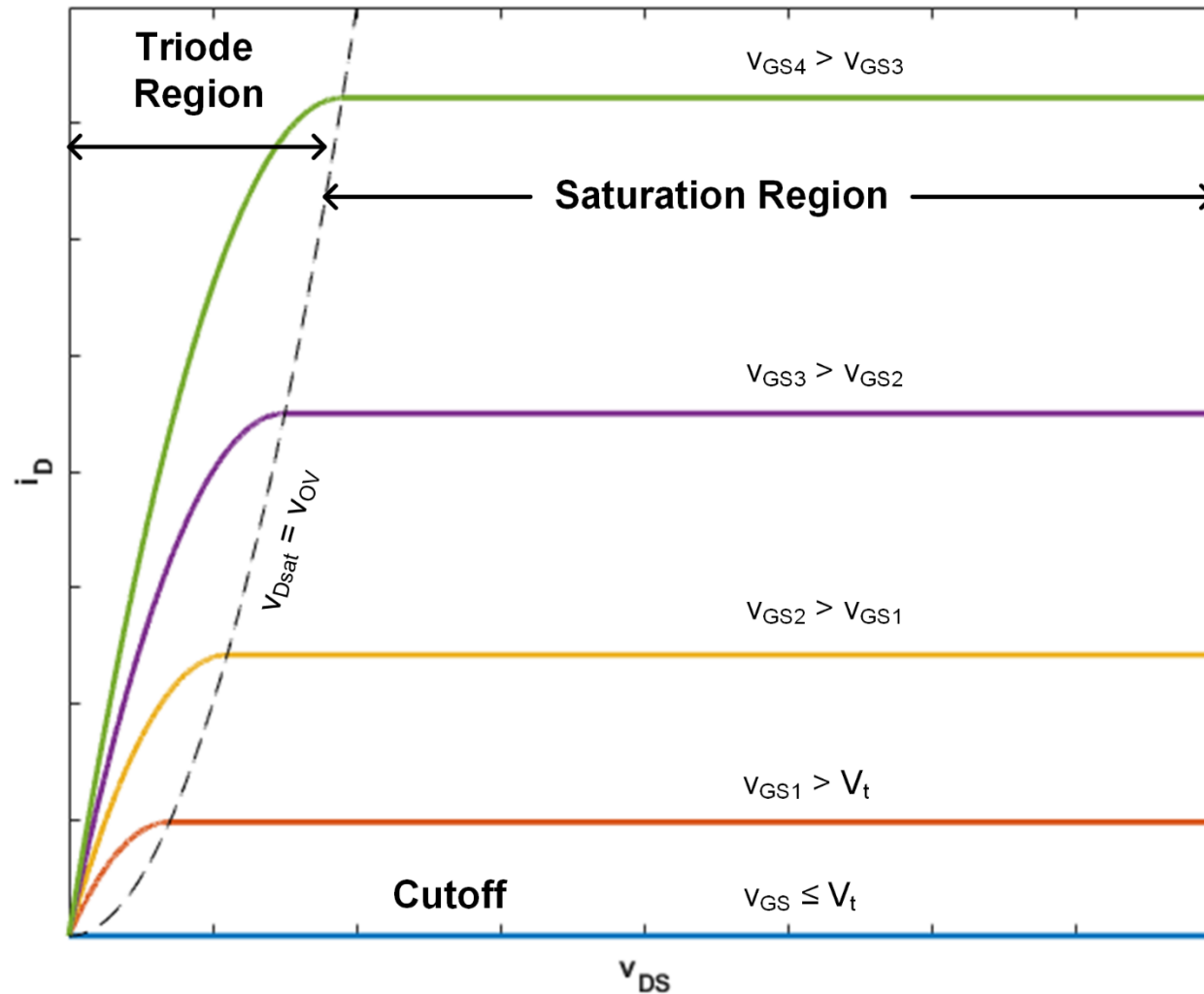
$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2$$

$$i_D = \frac{1}{2} k'_n \frac{W}{L} v_{OV}^2$$



Output I-V Characteristic

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NMOS Operating Regions – Summary

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□ Cutoff:

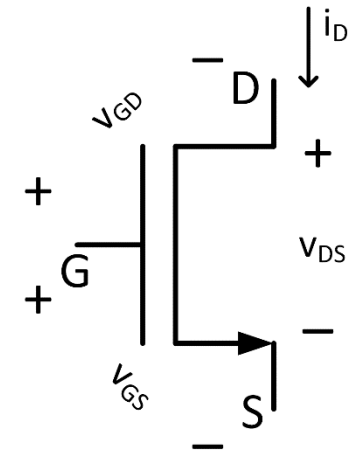
- $v_{GS} < V_t$
- $i_D = 0$

□ Triode:

- $v_{GS} > V_t$
- $v_{DS} < v_{OV}$ or $v_{GD} > V_t$
- $i_D = \mu_n C_{ox} \frac{W}{L} \left(v_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right) = k'_n \frac{W}{L} \left(v_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right)$

□ Saturation:

- $v_{GS} > V_t$
- $v_{DS} > v_{OV}$ or $v_{GD} < V_t$
- $i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_{tp})^2 = \frac{1}{2} k'_n \frac{W}{L} v_{OV}^2$



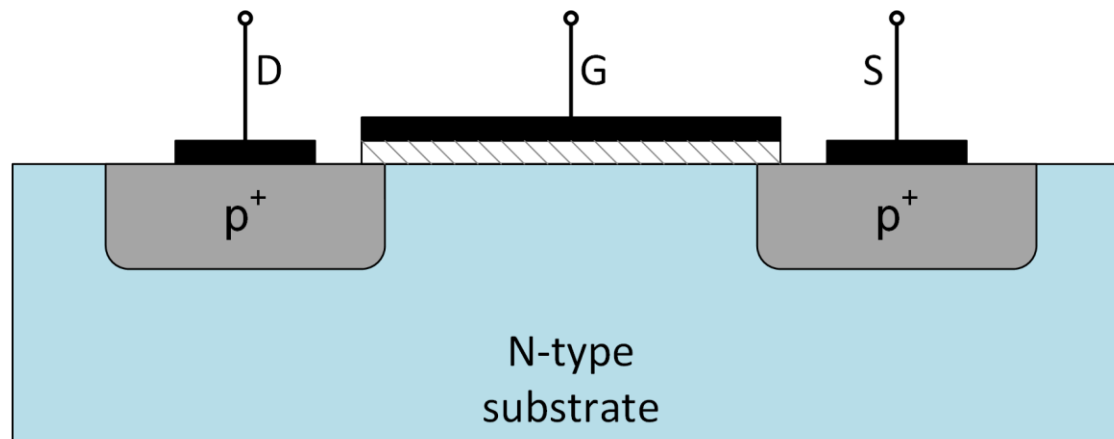
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P-Channel MOSFETs

P-Channel MOSFETs

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- Voltage polarities and doping types reversed relative to NMOS
 - ▣ N-type substrate
 - ▣ P⁺ drain and source
 - ▣ Negative threshold voltage: $V_{tp} < 0$
 - ▣ Negative overdrive voltage: $v_{OV} = v_{GS} - V_{tp} < 0$
 - ▣ Channel induced for $v_{GS} \leq V_{tp}$
 - ▣ Substrate connected to source or most positive circuit voltage



PMOS – Operating Regions

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□ Cutoff:

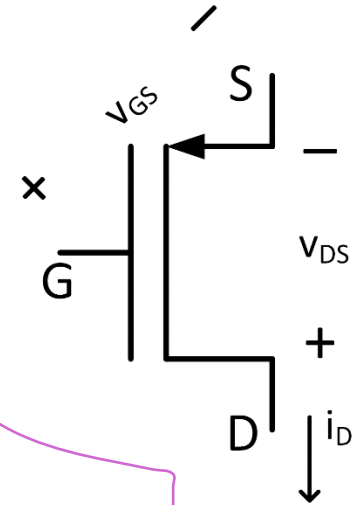
- $v_{GS} > V_{tp}$
- $i_D = 0$

□ Triode:

- $v_{GS} < V_{tp}, \quad |v_{GS}| > |V_{tp}|$
- $v_{DS} > v_{OV}, \quad |v_{DS}| < |v_{OV}|$
- $i_D = \mu_p C_{ox} \frac{W}{L} \left(v_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right) = k'_p \frac{W}{L} \left(v_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right)$

□ Saturation:

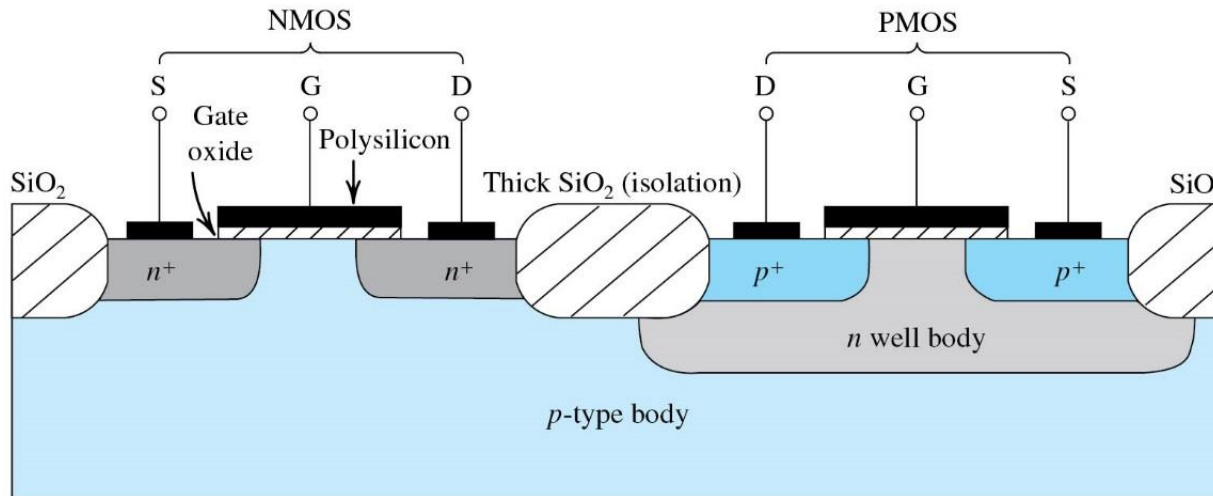
- $v_{GS} < V_{tp}, \quad |v_{GS}| > |V_{tp}|$
- $v_{DS} < v_{OV}, \quad |v_{DS}| > |v_{OV}|$
- $i_D = \frac{1}{2} k'_p \frac{W}{L} (v_{GS} - V_{tp})^2 = \frac{1}{2} k'_p \frac{W}{L} v_{OV}^2$



CMOS

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- Complementary MOS or **CMOS**
 - ▣ Both NMOS and PMOS fabricated on the same chip
- P-type substrate
- PMOS devices fabricated in ***n wells***
- Most modern MOS chips are fabricated using CMOS technology



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Large-Signal MOSFET Model

Equivalent Circuit Models

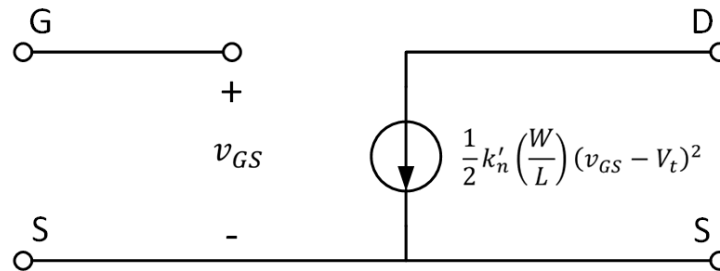
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- As was the case for BJTs, we use two types of equivalent-circuit models for MOSFETs:
- ***Large-signal model***
 - ▣ Models the transistor's behavior to DC signals
 - ▣ Used to determine the transistor's DC operating point
- ***Small-signal model***
 - ▣ Models the behavior in response to small signals
 - ▣ Describes the response to the AC signals to be amplified
 - ▣ Properties of the small-signal model determined by the DC operating point

Large-Signal Model – Saturation

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- Large-signal behavior in the saturation region is modeled by the following circuit:

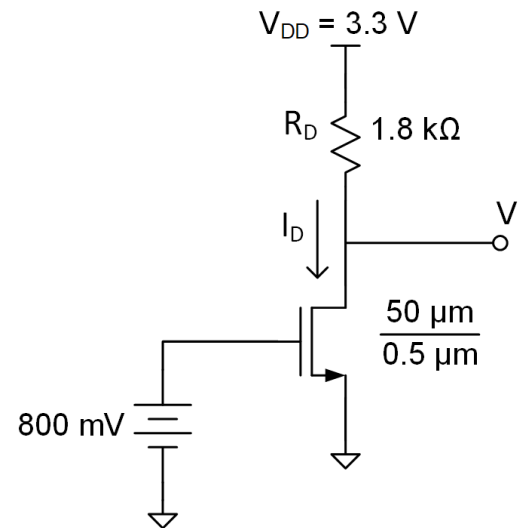


- Replace the transistor with the appropriate model to determine the DC operating point (Q-point)
- Saturation-region bias assumed
 - ▣ If incorrect, model will say otherwise
- **Because $i_G = 0$, we generally do not need to explicitly use the equivalent circuit model for large-signal analysis for MOSFETs**
 - ▣ Often just use the I-V model

DC Operating Point – Example 1

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- Determine I_D and V_D for the following circuit
 - ▣ Is the device operating in the saturation region?
- The process transconductance parameter:



$$\mu_n = 500 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

$$C_{ox} = 3.8 \frac{\text{fF}}{\mu\text{m}^2}$$

$$V_t = 500 \text{ mV}$$

$$L = 0.5 \mu\text{m}$$

$$W = 50 \mu\text{m}$$

$$k'_n = \mu_n C_{ox} = 500 \times 10^{-4} \frac{\text{m}^2}{\text{V} \cdot \text{s}} \cdot \frac{3.8 \times 10^{-15} \text{ F}}{1 \times 10^{-12} \text{ m}^2} = 190 \frac{\mu\text{A}}{\text{V}^2}$$

- Use the saturation region large-signal model
 - ▣ I-V model, not necessarily the equivalent-circuit model
 - ▣ Analysis will indicate if saturation assumption is incorrect

DC Operating Point – Example 1

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- Drain current in saturation:

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

$$I_D = \frac{1}{2} \cdot 190 \frac{\mu A}{V^2} \cdot \left(\frac{50 \mu m}{0.5 \mu m} \right) (800 mV - 500 mV)^2$$

$$I_D = 855 \mu A$$

- Voltage at the drain:

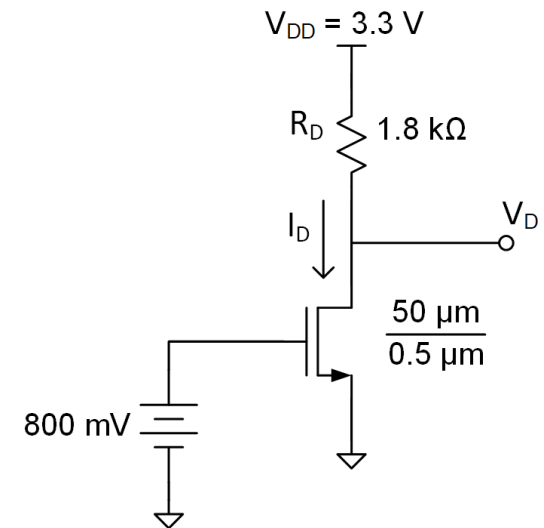
$$V_D = V_{DD} - I_D R_D$$

$$V_D = 3.3 V - 855 \mu A \cdot 1.8 k\Omega$$

$$V_D = 1.76 V$$

- The device is operating in the saturation region
 - ▣ The drain-to-source voltage exceeds the overdrive voltage

$$V_{DS} = V_D = 1.76 V > V_{OV} = V_{GS} - V_t = 300 mV$$



DC Operating Point – Example 2

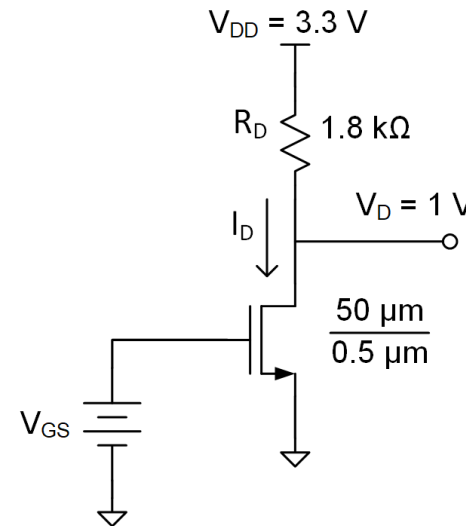
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- For the same circuit, determine V_{GS} for $V_D = 1\text{ V}$
 - Is the device still operating in the saturation region?

- For $V_D = 1\text{ V}$

$$I_D = \frac{V_{DD} - V_D}{R_D} = \frac{3.3\text{ V} - 1\text{ V}}{1.8\text{ k}\Omega}$$

$$I_D = \frac{2.3\text{ V}}{1.8\text{ k}\Omega} = 1.28\text{ mA}$$



$$\mu_n = 500 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

$$C_{ox} = 3.8 \frac{\text{fF}}{\mu\text{m}^2}$$

$$V_t = 500\text{ mV}$$

$$L = 0.5\text{ }\mu\text{m}$$

$$W = 50\text{ }\mu\text{m}$$

- Assuming saturation-region operation

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2 = \frac{1}{2} \cdot 190 \frac{\mu\text{A}}{\text{V}^2} \cdot \left(\frac{50\text{ }\mu\text{m}}{0.5\text{ }\mu\text{m}} \right) \cdot V_{OV}^2 = 1.28\text{ mA}$$

- Solving for the overdrive voltage

$$V_{OV} = 367\text{ mV}$$

DC Operating Point – Example 2

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- The required gate-to-source voltage is

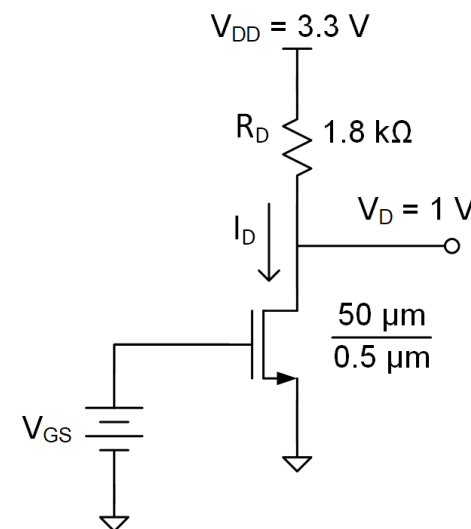
$$V_{GS} = V_t + V_{OV}$$

$$V_{GS} = 500 \text{ mV} + 367 \text{ mV}$$

$$V_{GS} = 867 \text{ mV}$$

- The drain-to-source voltage exceeds the overdrive voltage
 - ▣ The transistor is operating in the saturation region

$$V_{DS} = 1 \text{ V} > V_{OV} = 367 \text{ mV}$$



DC Operating Point – Example 3

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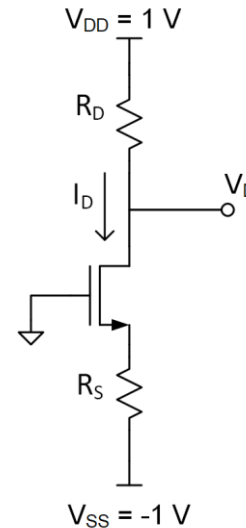
- Find R_D and R_S for

$$I_D = 200 \mu A \text{ and } V_D = 200 mV$$

- First, determine R_D

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{1 V - 200 mV}{200 \mu A}$$

$$R_D = 4 k\Omega$$



$$k'_n = 400 \frac{\mu A}{V^2}$$

$$V_t = 500 mV$$

$$L = 0.5 \mu m$$

$$W = 15 \mu m$$

- Drain current in saturation is given by

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2 \rightarrow V_{OV} = \sqrt{\frac{2 I_D L}{k'_n W}}$$

$$V_{OV} = \sqrt{\frac{2 \cdot 200 \mu A \cdot 0.5 \mu m}{400 \mu A/V^2 \cdot 15 \mu m}} = 182.6 mV$$

DC Operating Point – Example 3

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- The gate-to-source voltage is

$$V_{GS} = V_t + V_{OV} = 500 \text{ mV} + 182.6 \text{ mV}$$

$$V_{GS} = 683 \text{ mV}$$

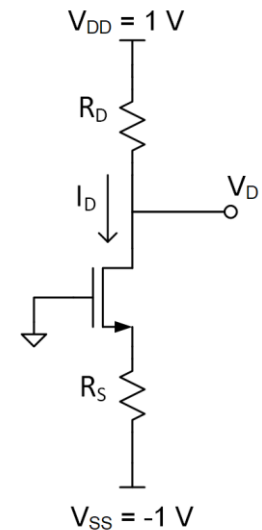
- The gate is grounded, so the source voltage is

$$V_S = V_G - V_{GS} = -683 \text{ mV}$$

- The source resistance is given by

$$R_S = \frac{V_S - V_{SS}}{I_D} = \frac{-683 \text{ mV} - (-1 \text{ V})}{200 \mu\text{A}}$$

$$R_S = 1.59 \text{ k}\Omega$$



DC Operating Point – Example 4

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- Find I_D and R_D for $V_D = 100 \text{ mV}$

- What is the drain-to-source resistance, r_{DS} ?

- The device is in the triode region:

$$V_{OV} = V_{GS} - V_t = 2 \text{ V} - 500 \text{ mV} = 1.5 \text{ V}$$

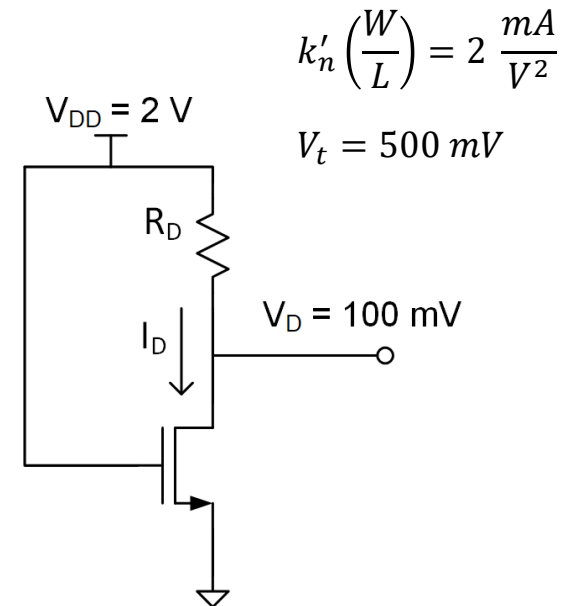
$$V_{DS} = 100 \text{ mV} < V_{OV}$$

- Drain current in triode is given by

$$I_D = k'_n \left(\frac{W}{L} \right) \left(V_{OV} - \frac{1}{2} V_{DS} \right) V_{DS}$$

$$I_D = 2 \frac{\text{mA}}{\text{V}^2} (1.5 \text{ V} - 50 \text{ mV}) 100 \text{ mV}$$

$$I_D = 290 \mu\text{A}$$



DC Operating Point – Example 4

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- The required drain resistance:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{1.9 \text{ V}}{290 \mu\text{A}}$$

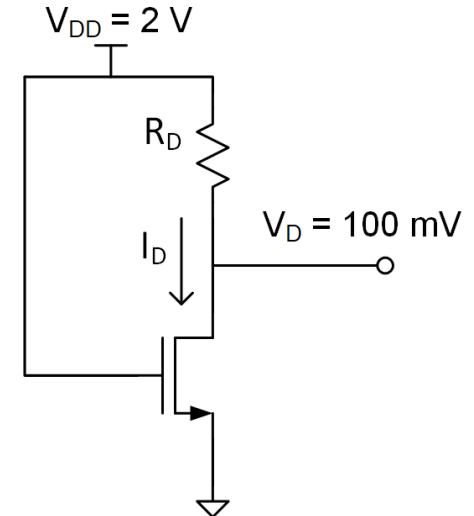
$$R_D = 6.55 \text{ k}\Omega$$

- The drain-to-source resistance in the triode region is given by:

$$r_{DS} = \frac{1}{k'_n \frac{W}{L} V_{OV}}$$

$$r_{DS} = \frac{1}{2 \frac{\text{mA}}{\text{V}^2} 1.5 \text{ V}}$$

$$r_{DS} = 333 \Omega$$



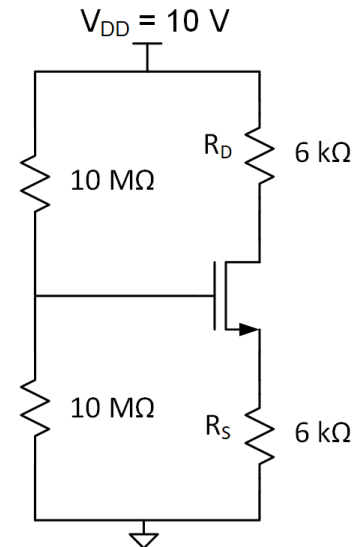
DC Operating Point – Example 5

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- Find I_D , V_G , V_D , and V_S for the following circuit
- The gate voltage is simply set by the voltage divider

$$V_G = 10\text{ V} \frac{10\text{ M}\Omega}{10\text{ M}\Omega + 10\text{ M}\Omega}$$

$V_G = 5\text{ V}$



$$k'_n \left(\frac{W}{L} \right) = 1 \frac{\text{mA}}{\text{V}^2}$$

$$V_t = 1\text{ V}$$

- Assuming operation in the saturation region, drain current is

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (V_{GS} - V_t)^2 = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (V_G - I_D R_S - V_t)^2$$

$$I_D = 0.5 \frac{\text{mA}}{\text{V}^2} (5\text{ V} - I_D R_S - 1\text{ V})^2$$

DC Operating Point – Example 5

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$$I_D = 0.5 \frac{mA}{V^2} (4 V - I_D \cdot 6 k\Omega)^2$$

- This is a quadratic equation for I_D

$$I_D = 18E3 I_D^2 - 24 I_D + 8E - 3$$

- Two possible solutions:

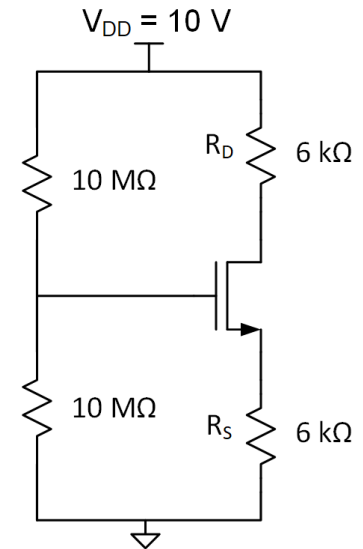
$$I_D = 889 \mu A \text{ or } I_D = 500 \mu A$$

- For $I_D = 889 \mu A$

$$V_D = V_{DD} - I_D R_D = 10 V - 889 \mu A \cdot 6 k\Omega = 4.67 V$$

$$V_S = I_D \cdot R_S = 889 \mu A \cdot 6 k\Omega = 5.33 V$$

$$V_{GS} = V_G - V_S = -333 mV$$



DC Operating Point – Example 5

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- For $I_D = 889 \mu A$
 - ▣ $V_{GS} < 0$
 - ▣ The transistor is in the cut-off region
- The valid solution to the quadratic equation must be

$$I_D = 500 \mu A$$

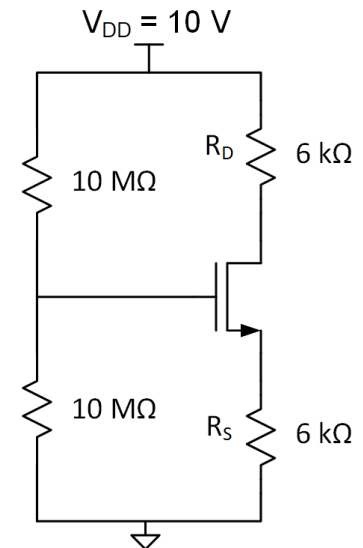
- The drain and source voltages are

$$V_D = V_{DD} - I_D R_D = 10 V - 500 \mu A \cdot 6 k\Omega$$

$$V_D = 7 V$$

$$V_S = I_D \cdot R_S = 500 \mu A \cdot 6 k\Omega$$

$$V_S = 3 V$$



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Channel-Length Modulation

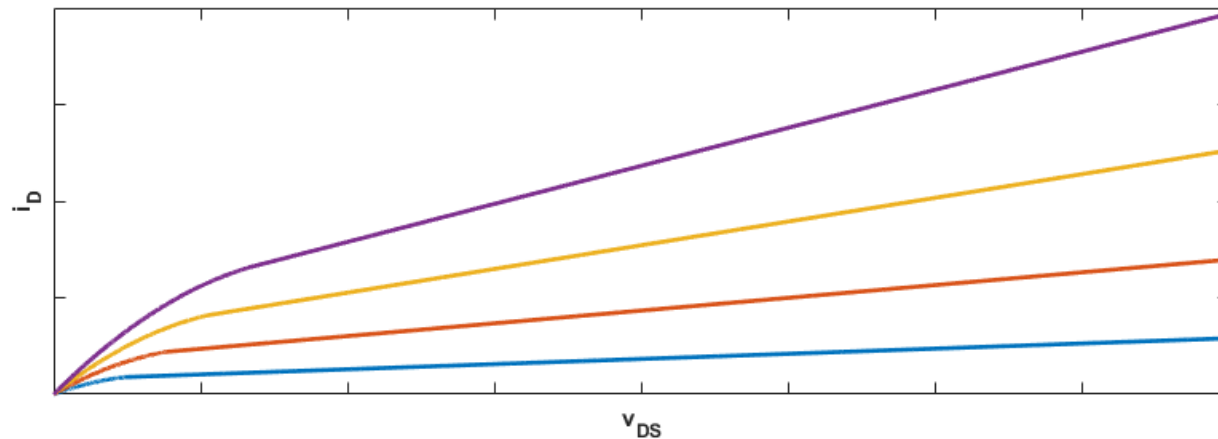
Channel-Length Modulation

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- So far, our MOSFET model in the saturation region models drain current as independent of v_{DS}

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2$$

- ▣ Flat I-V characteristic in the saturation region
- In reality, current increases as v_{DS} increases

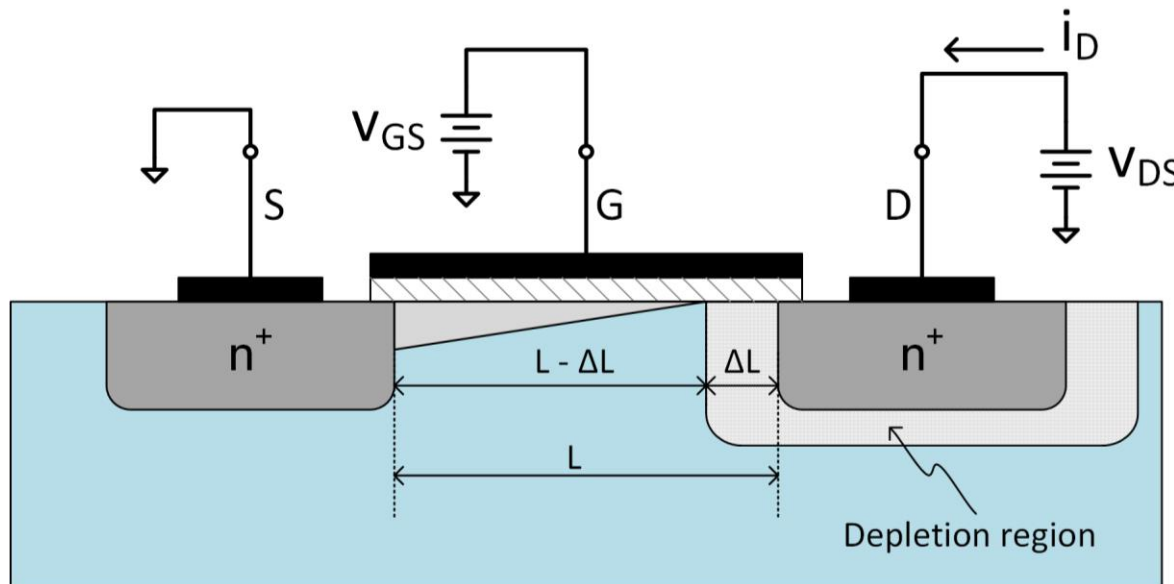


- Increase in i_D due to ***channel-length modulation***

Channel-Length Modulation

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- Justification for constant saturation current was:
 - ▣ Channel shape does not change after pinch-off occurs
 - ▣ Any additional v_{DS} is dropped across the depletion region surrounding the drain
- ***But, as v_{DS} increases, the drain depletion region increases, and the channel length decreases***



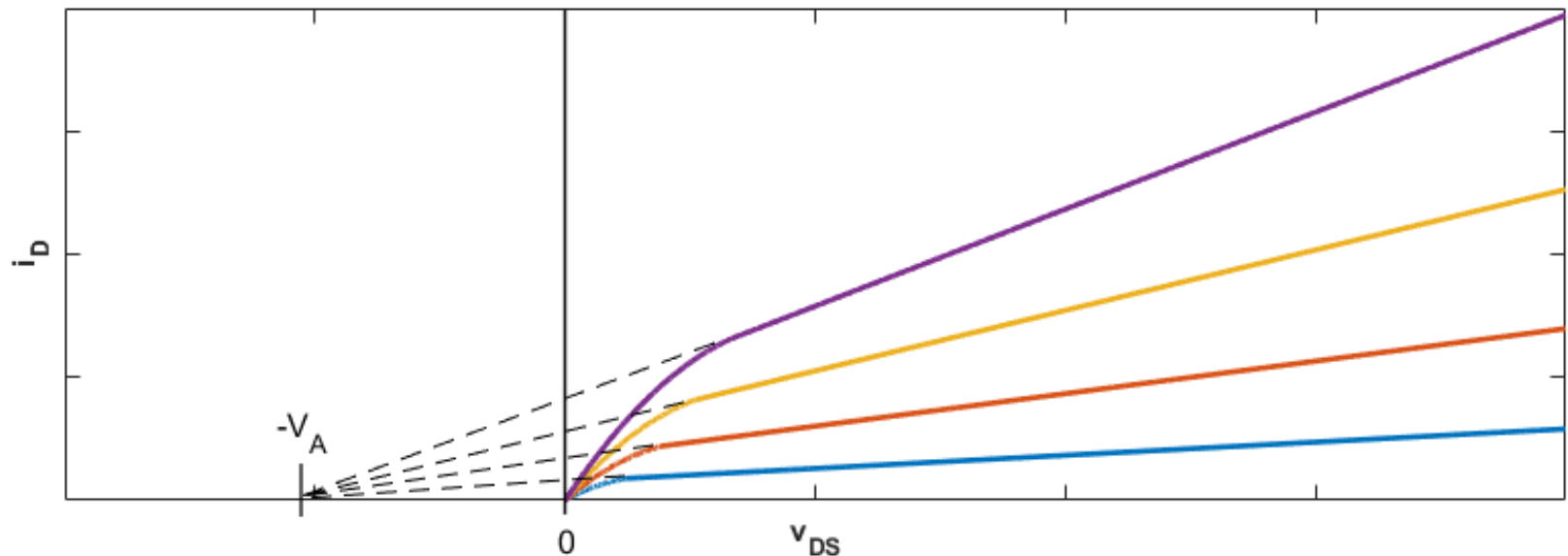
Channel-Length Modulation

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- Drain current is inversely proportional to channel length

$$i_D \propto \frac{1}{L}$$

- So, as v_{DS} increases, L decreases, and i_D increases
 - ▣ Non-zero slope in the saturation region of the i_D - v_{DS} curve:



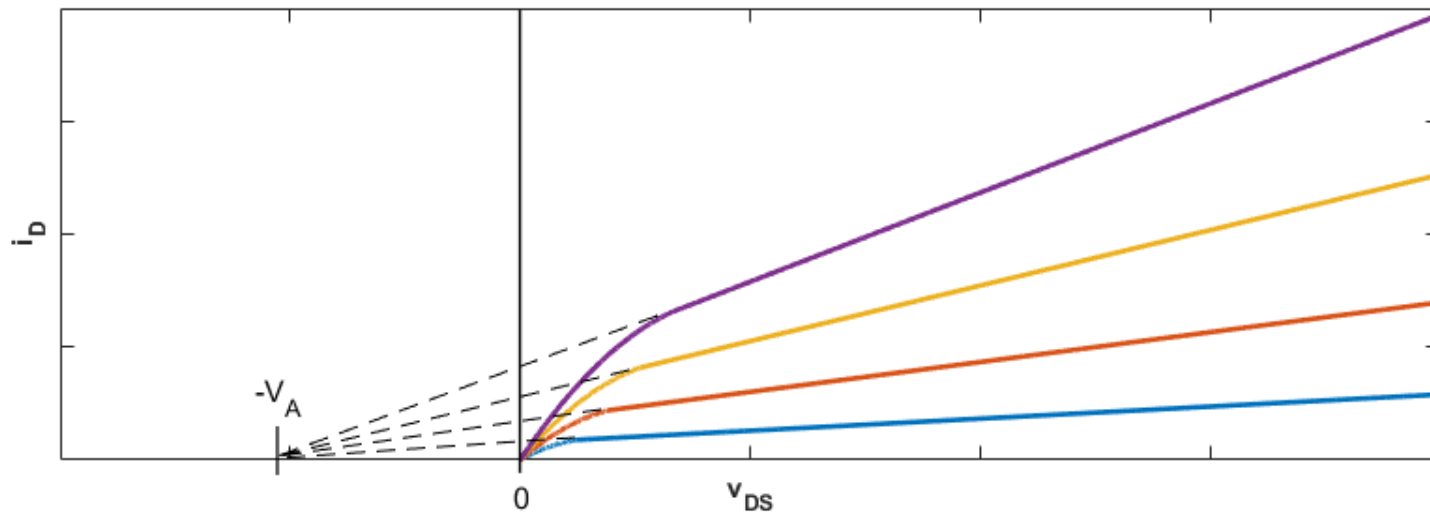
Channel-Length Modulation

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- This effect is accounted for by the **channel-length modulation parameter, λ**

$$i_D = \frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2 (1 + \lambda v_{DS})$$

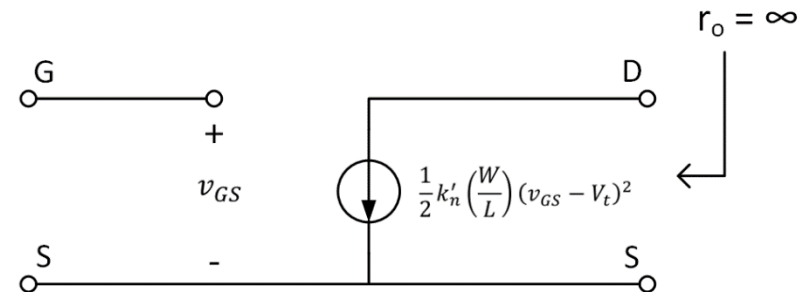
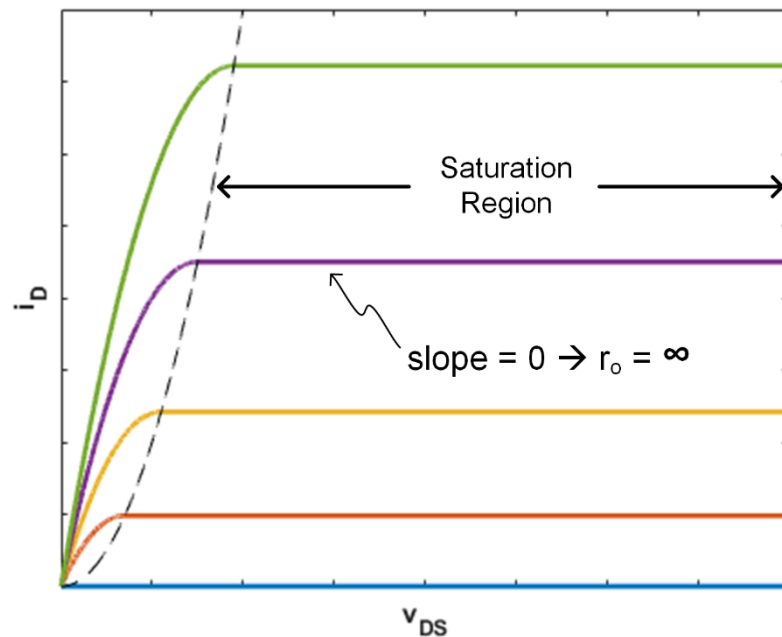
- ▣ λ is inversely proportional to channel length, L
- ▣ λ related to the **Early voltage**: $\lambda = \frac{1}{V_A}$



Output Resistance

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- Slope of the i_D - v_{DS} characteristic is the inverse of the transistor's **output resistance**
 - ▣ Resistance seen looking into the drain
- Constant saturation current implies **infinite output resistance**



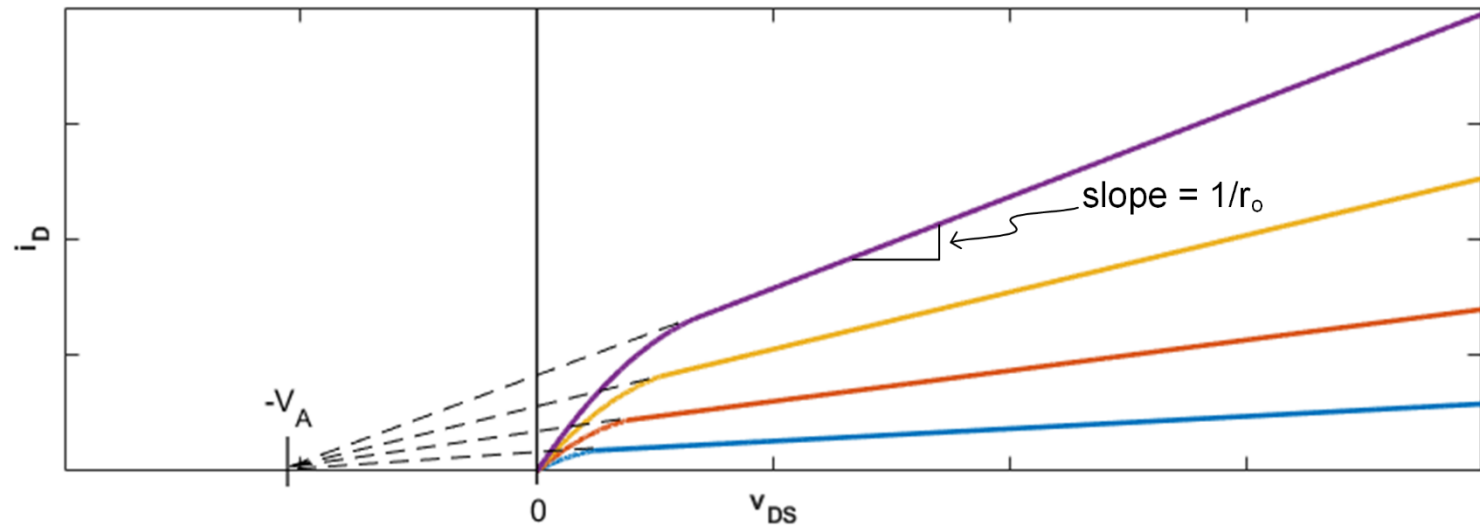
Output Resistance

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- Output resistance given by

$$r_o = \left[\frac{\partial i_D}{\partial v_{DS}} \right]^{-1}$$

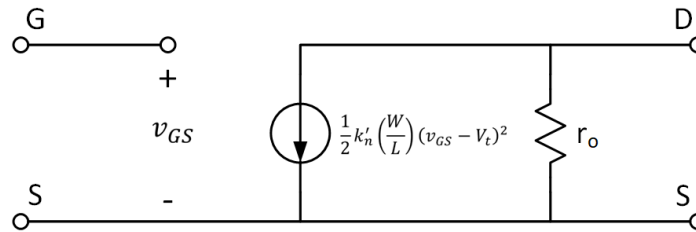
- ▣ The inverse of the slope of the i_D - v_{DS} characteristic
- Channel-length modulation results in finite output resistance



Output Resistance

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- Model the finite output resistance due to channel-length modulation by adding a resistor to our large-signal model



- Output resistance given by

$$r_o = \left[\frac{\partial i_D}{\partial v_{DS}} \right]^{-1} = \left[\frac{\partial}{\partial v_{DS}} \left(\frac{1}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2 (1 + \lambda v_{DS}) \right) \right]^{-1}$$

$$r_o = \left[\frac{\lambda}{2} k'_n \frac{W}{L} (v_{GS} - V_t)^2 \right]^{-1}$$

$$r_o = \frac{1}{\lambda I'_D} = \frac{V_A}{I'_D}$$

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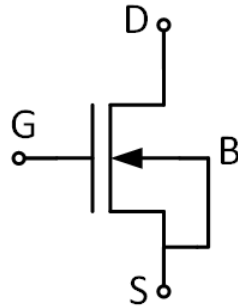
The Body Effect

The Body Effect

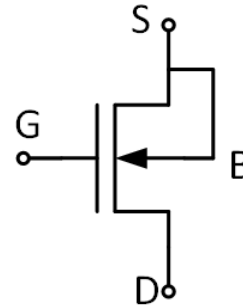
51

- So far, we have largely ignored the connection to the substrate
 - ▣ Equivalently, we have assumed it to be tied to the source:

NMOS:



PMOS:



- This a valid assumption for ***discrete*** devices
 - ▣ Not so for MOSFETs on integrated circuits (ICs)

The Body Effect

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- For integrated circuits, the substrate is typically tied to the most negative supply voltage for NMOS devices
 - PMOS n-wells tied to the most positive supply voltage
- Substrate for a given device may well be biased below its source voltage (above for PMOS)
 - This is the bias voltage for the channel region
- **For $V_{SB} > 0$ ($V_{SB} < 0$ for PMOS), the *threshold voltage is effectively increased***
 - This is the *body effect*

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MOSFETs as Switches

MOSFETs as Switches

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- Our focus is on the use of MOSFETs for ***linear amplifiers*** in ***analog circuits***
 - ▣ Operation in the ***saturation*** region
- MOSFETs are also useful as ***switches*** in ***digital circuits***
 - ▣ Microprocessors contain *billions* of MOSFETs used as switches on a single chip
- When operating as a switch, MOSFETs alternate between the ***triode*** (on, closed) and ***cutoff*** (off, open) regions

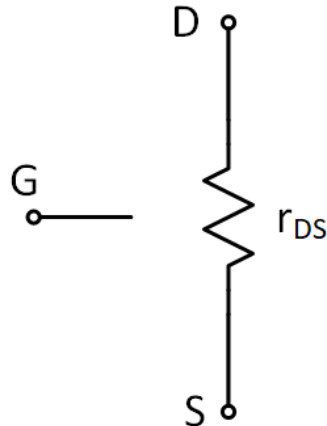
Triode/Cutoff Region Models

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- MOSFETs used as **switches** operate alternately in the **triode** (closed) and **cutoff** (open) regions
- Equivalent circuit models:

Triode Region (ON):

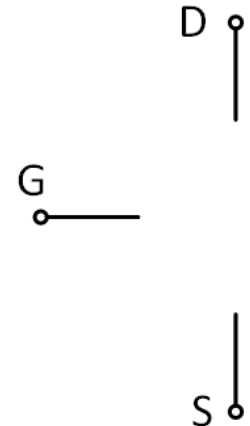
- $V_{GS} > V_t$
 - $V_{GS} = V_{DD}$



- Switch is on
- $I_D \geq 0$
- $V_{DS} = I_D r_{DS} < V_{OV}$

Cutoff Region (OFF):

- $V_{GS} < V_t$
 - $V_{GS} = 0$

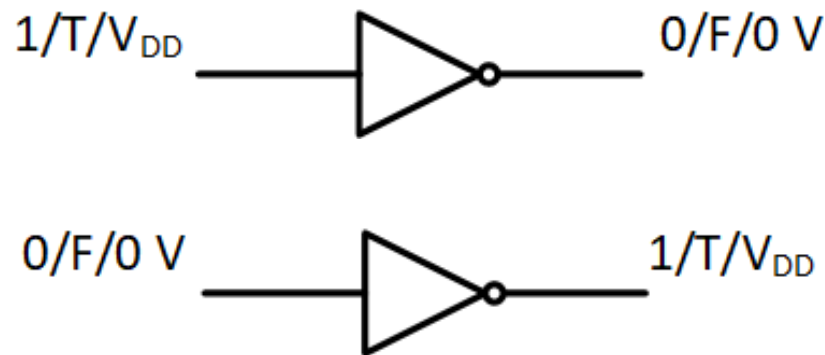


- Switch is off
- $I_D = 0$
- $V_{DS} = V_{DD}$

Inverters

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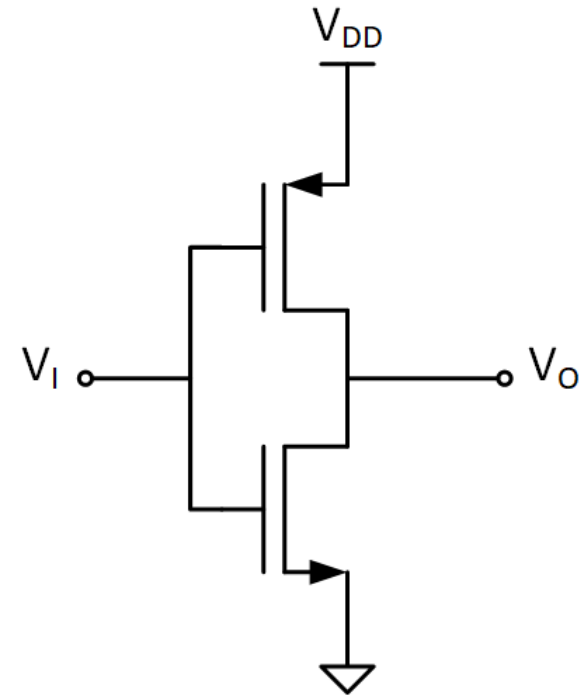
- The ***inverter*** is a fundamental building block of ***digital logic circuits***
- Output is the ***inverse*** of the input
 - When the input is a logic high (1/T) the output is low (0/F)
 - When the input is low (0/F) the output is high (1/T)



CMOS Inverter

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- CMOS inverters make use of NMOS and PMOS devices acting as switches
 - Input applied to gate of each device
 - Output taken from their drain terminals
 - When one switch is on, the other is off
 - Output connected to either V_{DD} or ground



CMOS Inverter

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□ Input is high:

- $V_I = V_{DD}$

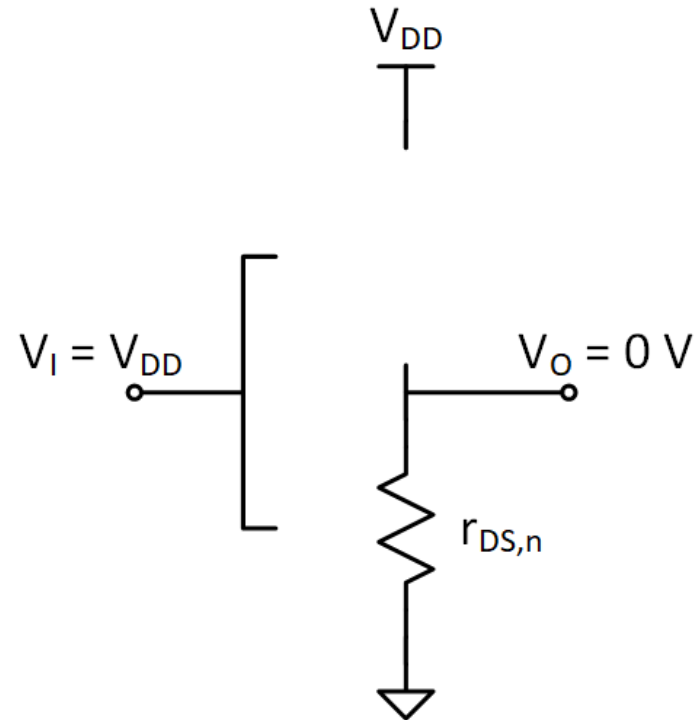
- $|V_{GS_p}| = 0\text{ V} < |V_{t_p}|$

- PMOS device is off

- $V_{GS_n} = V_{DD} > V_{t_n}$

- NMOS device is on

- $V_O \approx 0\text{ V}$



CMOS Inverter

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□ Input is low:

- $V_I = 0\text{ V}$

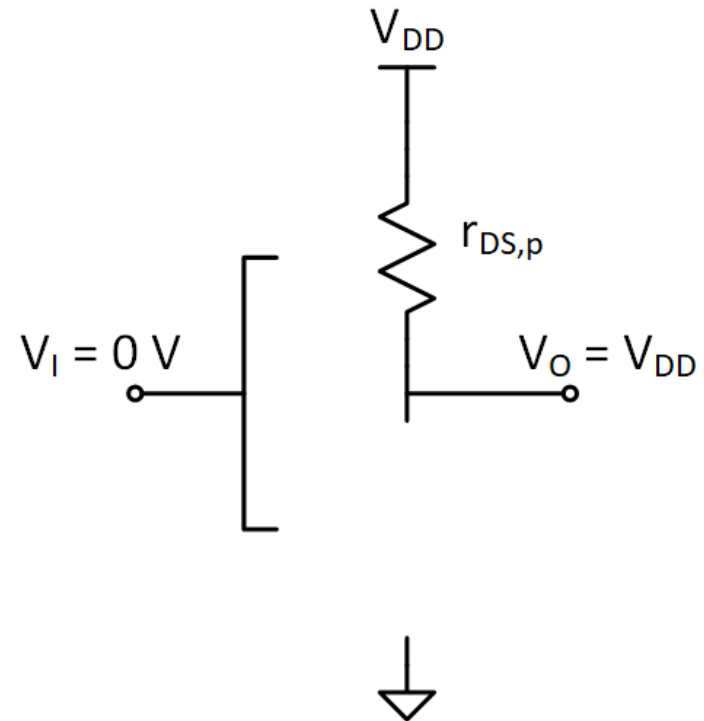
- $|V_{GS_p}| = V_{DD} > |V_{t_p}|$

- PMOS device is on

- $V_{GS_n} = 0\text{ V} < V_{t_n}$

- NMOS device is off

- $V_O \approx V_{DD}$



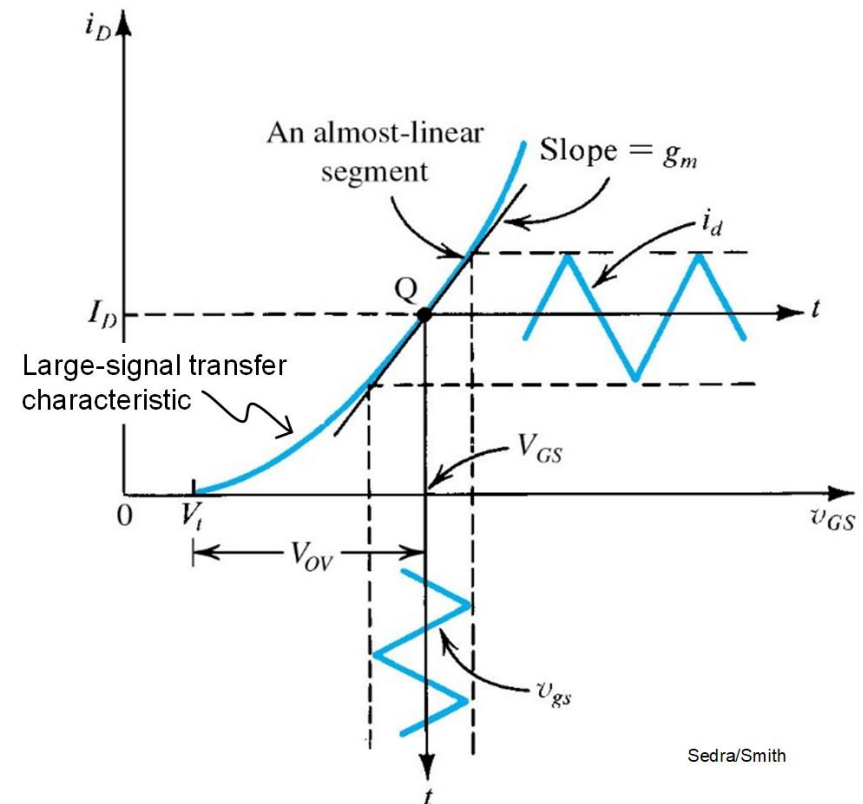
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Small-Signal Models

MOSFET Small-Signal Hybrid- π Model

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- Just as with BJTs, we use the large-signal model to determine the MOSFET's **DC operating point**
 - ▣ DC terminal voltages and drain current
- Need a **small-signal model** to describes the MOSFETs response to small signals
 - ▣ To describe its behavior as an amplifier
- Small-signal model parameters determined by the DC operating point



Sedra/Smith

MOSFET Small-Signal Hybrid- π Model

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- Similar to the BJT hybrid- π model
 - ▣ But, input resistance at the gate is infinite:

$$i_g = 0$$

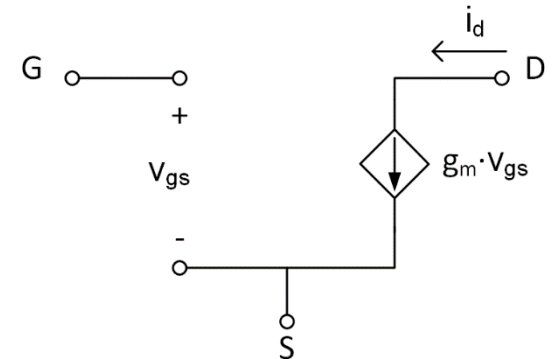
- **Transconductance**, g_m , defined as

$$g_m \equiv \frac{\partial i_d}{\partial v_{gs}}$$

- ▣ Where i_d and v_{gs} are the **small-signal components** of the drain current and gate-source voltage, respectively
- Transconductance determined by the DC operating point:

$$g_m = \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_t)$$

$$g_m = k'_n \left(\frac{W}{L} \right) V_{OV}$$



MOSFET Small-Signal Hybrid- π Model

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$$g_m = k'_n \left(\frac{W}{L} \right) V_{OV}$$

- Recall that DC drain current is given by

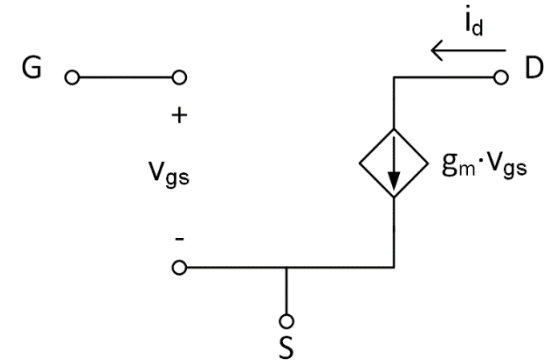
$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$$

- Solving for V_{OV} :

$$V_{OV} = \sqrt{\frac{2I_D}{k'_n \frac{W}{L}}}$$

- Substituting the V_{OV} expression into the g_m expression:

$$g_m = k'_n \left(\frac{W}{L} \right) \sqrt{\frac{2I_D}{k'_n \frac{W}{L}}} = \sqrt{2k'_n \left(\frac{W}{L} \right) I_D}$$



MOSFET Small-Signal T-Model

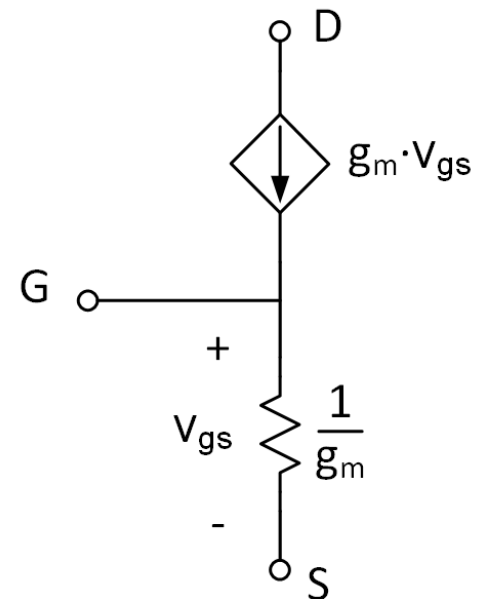
64

- An alternative small-signal MOSFET model is the T-model
 - ▣ As with the BJT T-model, useful when there is source resistance
- Though not immediately obvious, note that $i_g = 0$

$$i_d = i_s = g_m v_{gs}$$

- Resistance seen looking into the source:

$$R_s = \frac{1}{g_m}$$



MOSFET Small-Signal Models

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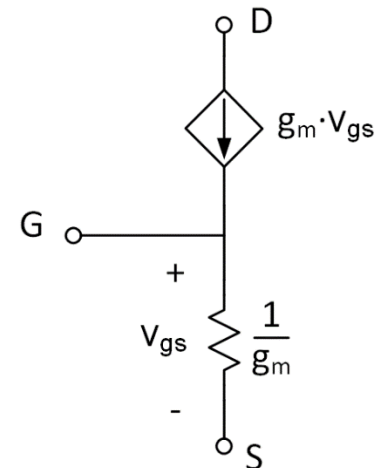
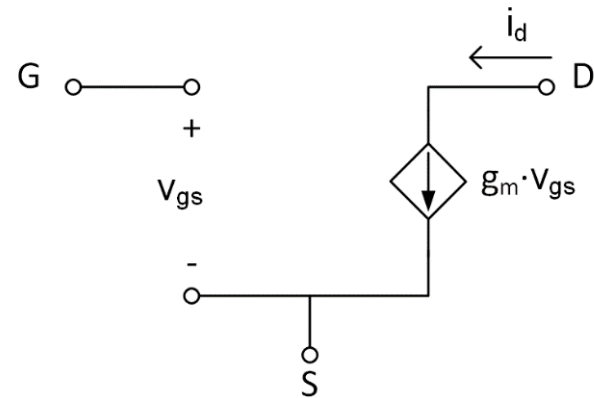
□ In summary:

- Transconductance in terms of overdrive voltage:

$$g_m = k'_n \left(\frac{W}{L} \right) V_{OV}$$

- Transconductance in terms of drain current:

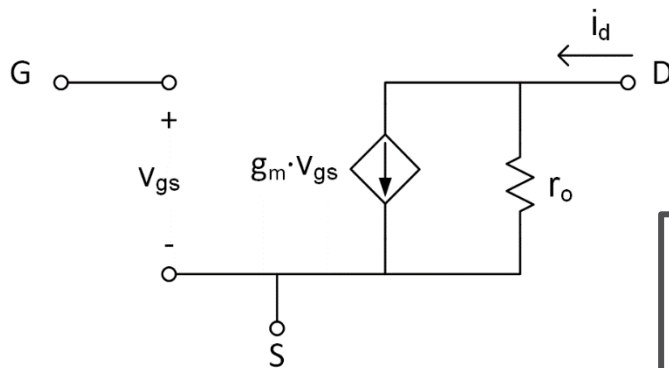
$$g_m = \sqrt{2k'_n \left(\frac{W}{L} \right) I_D}$$



Small-Signal Models – Finite r_o

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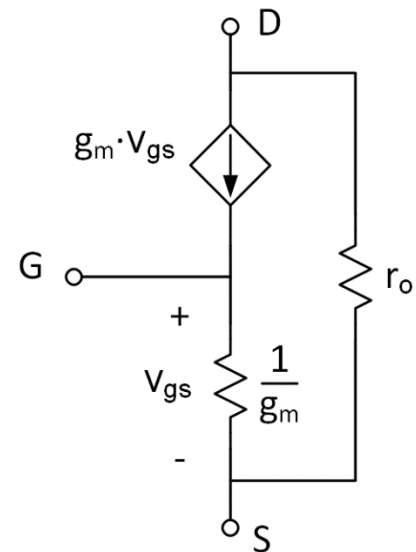
- We have seen that we can add output resistance to the large-signal model to account for channel-length modulation
 - Can do the same for the small-signal model
 - Will rarely, if ever, do this for the large signal model, but often will for the small-signal model



$$r_o = \frac{V_A}{I_D}$$

$$V_A = \frac{1}{\lambda}$$

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) V_{OV}^2$$



Using the MOSFET Models

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- In the next section of the course, we will look at the analysis and design of MOSFET amplifiers
 - Bias network and DC operating point
 - Signal-path
- As was the case for BJT amplifiers, our general procedure will be:
 - ***Large-signal analysis***
 - DC operating point
 - Small-signal model parameters
 - ***Small-signal analysis***
 - Circuit gain